Monitor Organization:	Monitor Department:	Monitor Title:
Quality, VP Office	*1500*	*RLPM Office Representative*

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RTX LIFECYCLE PROGRAM MANAGEMENT (RLPM) REVIEW PROCESS.

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1 Global Trade (GT)

1.1 This Procedure contains no technical data.

1.2 Documents containing technical data used to meet the requirements of this procedure must be controlled in accordance with all applicable IPEC requirements (refer to MMP 60 & MMP 375).

2 SCOPE

2.1 This Management Manual Procedure (MMP) defines the Gated reviews used at Pratt & Whitney Canada (P&WC) and is part of the framework that ensures P&WC complies to the requirements of P&W WP 1.5 "RTX Lifecycle Program Management" (RLPM).

2.2 This MMP defines the gated review process that is used to approve the progress of major product and service development projects. The intent of the process is to secure authorization to enter the next phase of the program. Authorization to proceed to the next phase is granted via the Gates following successful demonstration that:

1) The program is on track relative to its target metrics, and
2) Satisfactory plans are in place to mitigate any program risks (refer to EOP 168).

2.2.1 A summary of the program, market, business case, and technical issues is presented at the gated reviews following prior, more detailed technical (refer to EOP 103 Design Process and Design Reviews) and business (refer to MMP515) reviews.

This MMP defines six gate reviews and two sub-gate reviews across the IPMP used to solicit:

Gate A Strategy - authorizes permission to Study Concept

Gate B Submit/ Commit authorizes the submission of a technical and commercial proposal to the customer

Gate C Startup - authorizes the program to pursue the design and development of the product or service

Sub-gate C.1 Detailed Design - authorize the pursuit of development beyond PDR phase into detailed design. Sub-gate requirement will be subject to tailoring according to program archetypes as per section 7

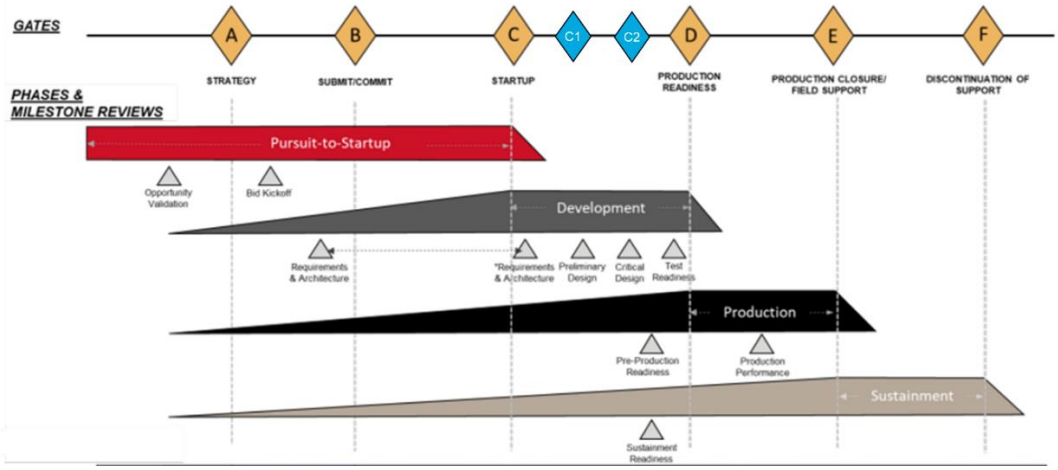
Sub-gate C.2 Production Hardware - authorize the ordering of production hardware and identification of any risk of ordering production hardware prior to completion of validation. Sub-gate requirement will be subject to tailoring according to program archetypes as per section 7.

Gate D Serial Production Readiness – Authorize the delivery of Serial production to the end customer

Gate E Production Closure/ **Field Support** - Review the product performance versus product requirements defined in earlier phases and capture program lessons learned for future programs or prepare for Production closure

Gate F Discontinuation of Support – agree on discontinuation of support and closure of RTX Lifecycle Program Management Process (RLPM)

2.3 The Gates and phases of the program are highlighted here below for illustration purposes. Lower-level reviews are for illustrative purposes only and might differ from one program to another.



2.4 This MMP applies to all product & service archetypes defined in section 7, this includes:

- New Engine, APU & IPPS Programs,
- Demonstrator Engine Programs,
- Derivative Models,

- Major Market Extension Programs, such as Supplemental Type Certification (STC) or Program Restarts
- All Industrial / Marine Applications,
- Major Scope Change, and
- Major Design Driven Cost Reduction Initiatives.

Note: The above includes military programs.

2.5 Program Scope Change is to be managed in accordance with EOP 174. Program Risk must be re-assessed with Scope Change (refer to EOP 168).

2.6 This MMP is aligned with the Integrated Product Management Process (MMP 219) as well as the Advanced Product Quality Planning (APQP) process, described in AS 9145.

2.7 The need for a Presidential RLPM Gate Review (PWPRES), conducted as per WP-1.5.1, is defined herein.

3 POLICY

3.1 The policy and methods, which appear in this procedure, apply throughout P&WC and its subsidiaries, divisions and other RTX business entities under its responsibility worldwide. Where used in this procedure, the term "the Company" applies to these subsidiaries, divisions and entities.

3.2 The policy and methods which appear in this procedure are part of the framework of P&W WP 1.5 "RTX Lifecycle Program Management" (RLPM).

3.3 To ensure alignment with the Companies strategic focus and resources, all activities must be authorized by the appropriate authorities within the Company, at pertinent stages of their implementation (called "gates"), through a defined process referred to as "RLPM Gates A-F", using mandatory checklists and templates.

4 OWNERSHIP AND APPROVAL

4.1 The Vice-President, Quality is the owner of this procedure. All interpretations and changes to policy require prior approval of the owner. Any questions regarding this MMP should be directed to GP PWC ePolicy (PWC.ePolicy@pwc.ca).

5 DEFINITIONS

5.1 GENERAL CONVENTIONS & DEFINITIONS

5.1.1 The following conventions are used in this procedure:

5.1.1.1 The words "shall", "will", or "must" indicate mandatory requirements.
5.1.1.2 The word "should" indicates a requirement with some flexibility allowed in compliance methodology. When choosing other methods to satisfy a "should" must be able to show that their approach meets the intent of the requirements of this standard.
5.1.1.3 Words "typical", "example", "for reference", or "e.g." indicate suggestions given for guidance only.
5.1.1.4 "Notes" are used for additional clarification.
5.1.1.5 Words or phrases with specific meaning pertaining to this document are defined below.
5.1.2 Entry Into Service (EIS) - The point in time when the first end user starts operating the product.
5.1.3 Major Scope Change - A significant change to the original engine requirements or definition during a development program that exceeds the approved AR investment, or $2 million USD on post certification programs.
5.1.4 Management reserve is defined as the investment and IRR contingencies allocated at the business case level. The Management reserve is a separate contingency, in addition to the specific provision for risks included by the business units on the various assumptions (NRE, FSC, DMC, other)

5.2 ACRONYMS

5.2.1	APQP	Advanced Product Quality Planning
5.2.2	APU	Auxiliary Power Unit
5.2.3	AR	Appropriation Request
5.2.4	CDN	Canadian Dollars
5.2.5	CDR	Critical Design Review
5.2.6	CGP	Controlled Goods Program
5.2.7	DR	Discrepancy Reports
5.2.8	EH&S	Environmental, Health & Safety
5.2.9	EIS	Entry into Service
5.2.10	FSC	Factory Standard Cost
5.2.11	FSW	Functional Standard work
5.2.12	GCP	RTX General Corporate Policy
5.2.13	IP	Intellectual Property
5.2.14	IPEC	Intellectual Property and Export Control
5.2.15	KPI	Key Performance Indicator
5.2.16	MPI	Marketing Proposal Inquiry
5.2.17	NPI	New Product Introduction
5.2.18	PDRS	Product Design Requirements Specification
5.2.19	PMMT	Program Model Management Team
5.2.20	PMT	Program Management Team
5.2.21	PPAP	RTX Production Part Approval Process
5.2.22	PWPres	Pratt & Whitney President Level RLPM Review
5.2.23	RLPM	RTX Lifecycle Program Management
5.2.24	STC	Supplemental Type Certification
5.2.25	MR	Management Reserve
5.2.26	SFV	Strategic Finance Valuation

5.3 RLPM GATE SCOPE AND REQUIREMENTS

Section 5.3 (in conjunction with PowerPoint templates listed in paragraph 8.1.110) defines the minimum RLPM Gate content requirements. If any of the mandatory material is omitted from the review, the reason of omission should be clearly documented using the appropriate tailoring checklist (see section 11) or substantiated in program plan.

5.3.1 GATE A - STRATEGY

This gate authorizes permission to Study Concept

Gate Trigger criteria

Gate A is triggered when an agreement is required to expend the resources required to complete an opportunity study beyond the level of resources committed following an Opportunity Validation Review

Gate Requirements

Gate A requirements include the following:

1. Pursue/No pursue decision
2. Program Management Team (PMT)/Capture team assigned
3. Baseline capture plan as part of the Program Plan
4. Establish requirement for partners participation
5. Draft Business Plan, as per section 6 of this document, highlighting
    - Marketing Segmentation and Environment
    - Opportunity Analysis completed
    - Marketing Strategy reviewed
6. Preliminary Development, Production, and Sustainment Plans as highlighted in Section 6 of this document
7. A Factory Standard Cost (FSC) study (refer to MMP 501) summarized in preliminary Production plan
8. A Preliminary Program plan as highlighted in section 6 of this document, containing
    - A review of Technical Configurations
    - A comprehensive assessment of the overall program risk associated with the enterprise program plan
    - Phase funding, schedule and resources/investments authorized
9. A product IP strategy plan including patent review schedule (refer to EOP 258)
10. The 'Enterprise KPI' Deliverable targets (refer to Form 11721)
11. A Global Trade review conducted and baseline Export/Import (EXIM) Plan, if required
12. IRs completed per plan
13. The sustainable engine strategy to meet company goals (refer to EOP 153)

14. The intent of the above requirements shall be met during gate review using standard Gate A template (form 11758), covering a requirement checklist, open actions, major changes since last gated review and identified exit criteria actions

Gate A is required to solicit agreement to expend the resources required to complete the study.

Gate A Authorization and Phase Closure
The amount of internal and external resources necessary as well as the budget to proceed to a customer proposal (e.g. through Gate B) must be determined during this gate.

5.3.2 GATE B SUBMIT/ COMMIT GATE

This gate authorizes the submission of a technical and commercial proposal to the customer

Gate Trigger criteria

This gate will be triggered to get approval for a commercial proposal to a customer or prior requesting an investment above the MMP 515 threshold (>$2M USD)

Gate Requirements

Gate B Requirements include the following:

1. A baseline Business plan (see section 6 of this document) highlighting
   - Market Summary including a competitive Analysis, Product positioning, Market outlook-volume
   - A summary of the business case (refer to MMP515) and contract deliverables
   - Required phase funding, schedule and resources/investments
   - Management Reserve identified per P&W WP 1.5 aligned with program risks identified (refer to MMP515 for standard work)
2. A Draft program Plan (see section 6 of this document) highlighting
   - A defined program scope including customer and internal requirements captured in a preliminary PDRS (see EOP 49)
   - Funded TRL/MRL maturation plan aligned with requirements in EOP 31 – Engineering Cost estimates (refer to MMP515 for standard work)
   - Labour requirements from all Business units to support program L1 plan, associated risks and critical skills identified.
   - Level 1 Plan (Creation of the proposed timeline including AR approval)
   - The intended partners and/or suppliers and the proposed quality plans as well as any additional company requirements

- The program archetype and tailoring review/approval for this program
- A Partnership/Collaboration strategy, including Partnership assessment (per MMP 507), workscope distribution (per EOP 102), and Program share allocation

3. A Commercial Response to Customer - OEM request overview, Commercial Opportunity Strategic Sandbox (COSS) and Commercial Response to Customer (Proposal Sandbox)
4. A summary of the technical proposal that was held prior to the Gate review (refer to EOP 60, 103 & PDD0250)
5. A comprehensive assessment of the overall program risk associated with the enterprise program plan, including:
   - All items identified as Red must have a mitigation plan presented (Including status of required resources financial and labour)
   - Identified low TRL and MRL (<6 items) identified and aligned with draft program plan TRL/MRL maturation plan
6. A Draft Production Plan (see section 6 of this document)
   - A review of the Factory Standard Cost (FSC)
   - A preliminary Industrialisation plan
   - Production Readiness Dashboard Exit Criteria (refer to IMP 07-01-01)
7. The 'Enterprise KPI' Deliverables (refer to Form 11721)
8. A Draft Sustainment plan (see section 6 of this document) highlighting the overall aftermarket strategy
9. A patent review (refer to EOP 258)
10. Global Trade (GT) assessment status and export licensing plan (Birth Certificate) (2)
11. Draft Development Plan (see section 6 of this document)
12. A program communication strategy
13. The program sustainability requirements for the complete product life cycle to meet company and customer goals
14. The intent of the above requirements shall be met during gate review using standard Gate B template (form 11760), using the gate requirement checklist (see section 11), open actions, major changes since last gated review and identified exit criteria actions

Gate B Authorization to proceed and Phase Closure

Authorization and commitment to proceed through gate B shall lead to the submission of a technical and commercial proposal to the customer. Upon Gate B approval, ensure AR is submitted in system for approval (as per MMP 515).

5.3.3 GATE C STARTUP

This gate authorizes the program to pursue the design and development of the product or service and initiate NRE

Gate Trigger criteria

Upon availability of the following but no later than 6 months after contract signature, unless otherwise approved in Program plan, and with sufficient time to support sub-gate C1 readiness and L1 milestones:

1. Product or service architecture external Customer requirements, agreed compliance/non-compliance matrix as per customer contract and expected regulatory requirements
2. An agreed and funded program resource plan & funded TRL/MRL strategy to support key identified product development risks
3. A baseline version of Program, development, Production and Sustainment plans as per section 6 of this document and relevant Procedures

Gate Requirements

Gate C Requirements will include the following.

1. A baselined program Plan (see section 6 of this document) highlighting
   - An agreed set of program goals
   - An approved Program resources; finances, people and critical skills, facilities/capital investment, and digital technology to launch full development
   - A status from Business units on labour requirements identified in Gate B to support program L1 plan, associated risks and mitigation plans
   - An agreed L1 Plan
   - A Product overview and an agreed architecture concept
   - A PMMT agreed PDRS (see EOP 49) containing regulatory requirements, key customer requirements and Internal requirements
   - Any identified non-compliances to PDRS necessary to meet program goals, agreed by PMMT and summarized in program plan
2. Independent reviews completed per program plan
3. An approved capital investment plan and NRE status
4. An updated Business plan (see section 6 of this document) highlighting
5. A Marketing outlook summary
   - A review of the business case summary (refer to MMP515) and update to Management reserve (as per IR review)
   - Key contract deliverables
   - Any change in scope impacting the AR or the business case
6. A baselined Production Plan (see section 6 of this document) highlighting
   - An initial assessment of facility, site, and infrastructure plans and requirements.
   - An internal initial production capacity plan and/or factory model with a plan to review maturity during technical reviews, including manufacturing automation plans.

- Sourcing and Manufacturing Strategy (including PPAP plan with a list of unique part numbers including mitigation for common hardware issues) (refer to QMSP 07-05-02),
- An initial Production Readiness Dashboard Exit Criteria (refer to IMP 07-01-01)
- A Factory Standard Cost (FSC) summary and plan

7. A Comprehensive assessment of the overall program risk associated with the enterprise program plan, including:
   - A Summary of the technical risk identified at the technical review that was held prior to the Gate review (refer to EOP 103 & PDD0250),
   - All items identified as Red must have a mitigation plan presented (Including status of required resources financial and labour)
8. TRL and MRL 6, whichever is greater. If this is not achievable
   - There must be a proposed TRL/MRL maturation plan to result in achieving TRL level 6 by Preliminary Design Review (PDR) and an MRL 6 by Critical Design Review (CDR).
   - Funding of the TRL and MRL maturation plan must occur and include back up plans if unable to achieve the proposed TRL/MRL 6 level. TRL/MRL maturation plan shall be aligned with requirements in EOP 31 – Engineering Cost estimates and MMP515 guidance
   - The back up plans must meet both the Program and Business objectives.
9. A baselined Development Plan (see section 6 of this document) that defines the technical execution strategy
   - A summary of the preliminary Verification and Validation (V&V) plan highlighting key methods identified associated with major investments (ref. EOP 60 for process and validation methodologies)
   - As a function of partnership risk profile, and assigned scope, a dedicated partnership development plan may be required
10. Global ITC assessment status and export licensing given the sourcing plan and scope changes
11. A patent review (refer to EOP 258),
12. A baselined Sustainment Plan (see section 6 of this document) that highlights the overall Aftermarket strategy
13. A review of the sustainability and Environment, Health & Safety (EH&S) metrics for the complete product life cycle
14. The 'Enterprise KPI' Deliverables (see Form 11721),

The intent of the above requirements shall be met during gate review using standard Gate C template (form 11760), using the gate requirement checklist (see section 11), open actions, major changes since last gated review and identified exit criteria actions

Specific tailoring for Gate Requirements can be found in section 6 for each archetype. Additional program tailoring for archetypes defined in section 6 will be defined in the Program management plan as per section 7 of the current document.

Gate C Authorization to proceed and Phase Closure

Gate C authorizes the continuation of the program towards a PDR and detailed design phase.

5.3.4 SUB-GATE C.1 DETAILED DESIGN

Sub Gate C.1 will authorize the program to continue development into detailed design.

Gate Trigger criteria

Sub-gate C.1 will be triggered following a successful completion of a PDR as per EOP 103. Sub-gate requirement will be subject to tailoring according to program archetypes as per section 7.

Sub-Gate Requirements

Sub-Gate C.1 Requirements and review material are similar to gate C. An update to all plans will be required at this stage as per Program management plan and following relevant BU procedures.

1. An updated program Plan (see section 6 of this document) highlighting
   - An update to program goals highlighting any differences since Gate C
   - An update to L1 Plan
   - A Product overview and an architecture concept following PDR
   - Any changes and compliance to documented regulatory requirements, customer or internal requirements (PDRS see EOP 49)
   - A status of compliance against PDRS including a PMMT agreed set of non-compliances to PDRS requirements
2. Independent reviews completed per program plan
3. An update to NRE status
4. An updated Business plan (see section 6 of this document) highlighting
   - A Marketing outlook summary
   - A review of the business case summary (refer to MMP515) and required update to Management reserve (as per IR review)
   - Status to key contract deliverables
   - Any change in scope impacting the AR or the business case
5. An updated Production Plan (see section 6 of this document) highlighting

- An updated assessment of facility, site, and infrastructure plans and requirements.
- An updated production capacity plan and/or factory model with a plan to review maturity during technical reviews, including manufacturing automation plans.
- An update to Sourcing and Manufacturing Strategy (including PPAP plan with a list of unique part numbers including mitigation for common hardware issues) (refer to QMSP 07-05-02),
- An update to Production Readiness Dashboard Exit Criteria (refer to IMP 07-01-01)
- A Factory Standard Cost (FSC) summary and plan

6. A Comprehensive assessment of the overall program risk associated with the enterprise program plan, including:
   - A Summary of the technical risk identified at the technical review (PDR) that was held prior to the gate review (refer to EOP 103 & PDD0250),
   - All items identified as Red must have a mitigation plan presented (Including status of required resources financial and labour)
7. TRL level 6 for those items identified in Gate C requiring TRL progression. MRL 6 for all components, if this is not achievable
   - There must be a proposed TRL/MRL maturation plan to result in achieving the MRL 6 by sub-gate C.2
   - Funding of the TRL and MRL maturation plan must occur and include back up plans if unable to achieve the proposed TRL/MRL 6 level. TRL/MRL maturation plan shall be aligned with requirements in EOP 31 – Engineering Cost estimates and MMP515 guidance
   - The back up plans must meet both the Program and Business objectives.
8. An update to Development Plan (see section 6 of this document) that defines the technical execution strategy
9. Global ITC assessment status and export licensing given the sourcing plan and scope changes
10. A patent review (refer to EOP 258),
11. A baselined Sustainment Plan (see section 6 of this document) that highlights the overall Aftermarket strategy
12. A review of the sustainability and Environment, Health & Safety (EH&S) metrics for the complete product life cycle
13. The 'Enterprise KPI' Deliverables (see Form 11721),
14. The intent of the above requirements shall be met during gate review using standard sub-Gate C.1 template (form 11760), using the gate requirement checklist (see section 11) , open actions, major changes since last gated review and identified exit criteria actions

5.3.5 SUB-GATE C.2 – PERMISSION TO ORDER PRODUCTION HARDWARE

Sub Gate C.2 will authorize the program to pursue ordering production hardware upon a successful review of main program risks

Gate Trigger criteria

Gate C.2 shall be called to ensure pre-production or production engine hardware provisioning inventory can be authorized within guidance here below while ensuring appropriate program readiness to achieve customer schedule commitments.

The Program plan should set a Gate C.2 date referencing the criteria below:

- After a successful Program CDR, after majority of development testing has been completed and aligned with BOM Change Control date set by the PMMT
- In time for long leadtime for ordering production HW at risk within guidance limit of < 5M USD
- The above not to be inside 18months leadtime of the first revenue service production engine delivery

The program plan shall be updated at Gate C.1 (see section 6) with a proposed date for gate C.2

Sub-Gate Requirements

Sub-gate C.2 requirements and review will include:

1. An updated program Plan (see section 6 of this document) highlighting
   - An update on the program status (strategy, marketing, economic & aftermarket),
   - An updated set of program goals including customer and internal requirements
   - An update of the L1 Plan
   - A review of product performance update based on preliminary verification of the functionality of the product, shortfalls and roadmap to meet KPIs
   - An update to compliance status against the PDRS (refer to EOP 49)
2. An update on NRE status
3. An updated Business plan (see section 6 of this document) highlighting
   - An update marketing outlook summary
   - A review of the business case summary (refer to MMP515) and required update to Management reserve (as per IR review)
   - Status to key contract deliverables
   - Any change in scope impacting the AR or the business case
4. An update to the Production Plan (see section 6 of this document) highlighting

- An update to Sourcing and Manufacturing Strategy (including PPAP plan - (refer to QMSP 07-05-02) including the ramp up to manufacturing plan to support EIP/EIS
- An update to the Production Readiness Dashboard Exit (refer to IMP 07-01-01)
- An updated status on Factory Standard Cost (FSC) summary and plan
- A review of the supply chain preparatory work as outlined in ASQR-01,

5. A Comprehensive assessment of the overall program risk associated with the enterprise program plan, including:

- Any risk of ordering production hardware prior to completion of validation,
- Any retrofit risks including proposed mitigation plans,
- All items identified as Red must have a mitigation plan presented (Including status of required resources financial and labor)

6. TRL and MRL > 6 of the product
7. An update to Development Plan (see section 6 of this document)
8. Conversion of the engineering BOM (eBOM) into manufacturing BOM (mBOM) or agreed PMMT plan for change control
9. Global ITC assessment status and export licensing given the sourcing plan and scope changes
10. A patent review (refer to EOP 258),
11. An update to Sustainment Plan (see section 6 of this document) highlighting the following
    - The aftermarket readiness dashboard of key deliverables to support service infrastructure
    - key deliverables to support service infrastructure and support (including ICA readiness plan)
12. A review of the sustainability and Environment, Health & Safety (EH&S) metrics for the complete product life cycle
13. The 'Enterprise KPI' Deliverables (see Form 11721),
14. The intent of the above requirements shall be met during gate review using standard subgate C.2 template (form 11760), using the gate requirement checklist (see section 11) , open actions, major changes since last gated review and identified exit criteria actions

5.3.6 GATE D – PERMISSION TO DELIVER SERIALISED PRODUCTION

Gate D authorizes the start of serialized production of the engine/service.

Gate D Trigger criteria

The Program plan shall set a Gate D date referencing the criteria below.

- Can be after delivery of pre-production engines planned in program to be returned for upgrade

- Can be after delivery of pre production engines planned to be upgraded at OEM but shall not exceed 20 units
- Should occur prior to manufacturing approval and 1st production delivery
- Timing shall be set to ensure meaningful assessment of industrializing maturation plan in preparation for the Product Performance Milestone and EIS readiness deliverables

The program plan shall be updated at Gate C.2 (see section 6) with a proposed date for gate D

Gate Requirements

Gate D Requirements and gated review content includes:

1. An updated program Plan (see section 6 of this document) highlighting
   - An update on the program status (strategy, marketing, economic & aftermarket)
   - Approved Program resources (finances, people, facilities/capital investment, and digital technology) to launch serialized production
   - An update of the L1 Plan
   - The 'enterprise KPI' Deliverables (see form 11721)
   - Successful completion of or plan for completion of V&V of component, product, and system level requirements with customer approval documented.
   - A compliance status against the PDRS (refer to EOP 49) where required with a time-bound, resourced and funded closure plan or associated risks reflected in enterprise program risk cube
   - Identification of funded program improvement plans based on customer/airframe Market Feedback Analysis, a Customer Oriented Results & Excellence (CORE) method, as required
   - A Product performance and architecture stability overview (including proof of manufacturing with lessons learned incorporated into design or manufacturing documentation)
   - The release both the Technical Data Package (TDP - including definition, ESA and SFI approvals) and Manufacturing Data Package (MDP - including MOS, AFS and approved repairs).
2. Conversion of the engineering BOM (eBOM) into manufacturing BOM (mBOM).
3. An updated development plan (see section 6 of this document) highlighting
   - Completion of all qualification testing and completion of all certification reports with remaining open items identified
   - Completion of all verification activities for first production hardware (reference EOP 57) including PFMEA, DFMEA, FAIR.
   - Status of compliance against certification requirements
4. An updated Business plan (see section 6 of this document) highlighting
   - An update marketing outlook summary

- A review of the business case summary (refer to MMP515) and required update to Management reserve (as per IR review)
- An updated status on key contract deliverables
- Any change in scope impacting the AR or the business case

5. An update to the Production Plan (see section 6 of this document) highlighting
   - Released production BOM and production sources identified for all parts with approved rate plans.
   - Gap assessment results of facility, site, and infrastructure plans and requirements in support of serialized production
   - A Factory Standard Cost (FSC) summary and plan established and funded
   - An updated Production Readiness Dashboard Exit Criteria (refer to IMP 07-01-01)
   - Industrial Strategy (including PPAP plan status as per QMSP 07-05-02) with resources in place (to include but not limited to staffing, equipment, tooling, and fixtures)
   - Documentation of Industrial risks and technical debt at Gate D with funded mitigation plans in place and communication to the customer as required
   - A list of identified post certification changes at gate D that require customer or field retrofit (OEUP) and consequential financial impact to business plan assumptions
6. Verification that all DR's having Hazardous Engine Condition or EH&S implications are closed with corrective action (refer to EOP 158), and evidence that all other DR have action plans
7. A Comprehensive assessment of the overall program risk associated with the enterprise program plan, including:
   - A summary of the Industrialization and technical risk identified at the technical review that was held prior to the Gated review (refer to EOP 103 & PDD0250),
   - All items identified as Red must have a mitigation plan presented (Including status of required resources financial and labor)
   - Identified risks requiring fleet data input (EIS Excellence plan - including MRO or field inspections or DPHM data to support ICA completion)
8. TRL 8 and MRL 9, whichever is greater. If this is not achievable
   - There must be a proposed TRL/MRL maturation plan to result in achieving TRL level 8 and MRL 9 by Entry into Production (EIP).
   - Funding and resource allocation plan of the TRL and MRL maturation plan must occur and include back up plans if unable to achieve the proposed TRL 8 and MRL 9 level.
   - The back up plans must meet both the Program and Business objectives.
9. Global Trade assessment status and export licensing plan (Birth Certificate) given the sourcing plan and scope changes
10. A patent review (refer to EOP 258), along with documentation Legal/Contracts/Compliance of program developed IP including sustainment strategies
11. NRE status

12. An update to Sustainment Plan (see section 6 of this document) highlighting the following
    - Completion of the sustainment customer support strategy plan (covering equipment, manuals, repair strategy, and training)
    - A review of the sustainability and Environment, Health & Safety (EH&S) metrics for the complete product life cycle
13. The intent of the above requirements shall be met during gate review using standard subgate D template (form 11760), using the gate requirement checklist (see section 11) , open actions, major changes since last gated review and identified exit criteria actions

Gate D Authorization and Phase Closure

The review will include a proposed maturity threshold with an upper limit that will determine the timing for conducting of Gate E.

Note: The authorization to deliver production engines is granted by the appropriate regulatory body. (refer to QMSP 10-06).

5.3.7 GATE E – FIELD SUPPORT AND PRODUCTION CLOSURE GATE

Gate E is called to review product performance versus product requirements defined in earlier phases in order to ensure lessons learned are captured in standard work for future programs

Gate E Trigger criteria

Gate E review is called upon sufficient gathering of field and production data, as defined in the Gate D Closure and to be captured in program plan (typically from 50 -100 engines delivered, or 24 months following Entry into Service (EIS).

Gate E is also triggered if production closure is required prior to reaching a minimum dormant period of 2 yrs since last production manufacturing activity. All requirements with "if applicable" here below are only used in the event that the gate is triggered to support production closure.

Gate Requirements

1. An updated program Plan (see section 6 of this document) highlighting
    - A compliance to program goals including customer and internal requirements (identified gaps to PDRS)
    - An update of the L1 Plan

- Identification of funded program improvement plans based on customer/airframe Market Feedback Analysis, a Customer Oriented Results & Excellence (CORE) method, as required

2. An updated Business Plan (see section 6 of this document) highlighting
   - A review of the business case summary (refer to MMP515) and required update to Management reserve (as per IR review)
   - Performance against contractual obligation assessment to determine customer and business case impact
3. A Comprehensive assessment of the overall program risk associated with the enterprise program plan, including:
   - A Summary of the technical risk identified at the technical review that was held prior to the Gate review (refer to EOP 103 & PDD0250),
   - All items identified as Red must have a mitigation plan presented and associated post certification investment financial resources (technical debt) required (aligned with AR)
   - Identified risks requiring fleet data input (EIS Excellence plan - including MRO or field inspections or DPHM data to support ICA completion)
4. Capture and communicate enterprise lessons learned for future programs
5. An update to the Production Plan (see section 6 of this document) highlighting
   - An update to the manufacturing resources (such as staff and equipment) plan
   - An update on Factory standard Cost
   - If applicable a plan for all government or customer furnished/owned property
   - If applicable, completion of all production contractual requirements
   - Industrialization Status, Production Readiness Dashboard Exit Criteria (refer to IMP 07-01-01)
   - Completion of MRL assessments (demonstrate that product is MRL9, show MRL readiness plan for new product changes)
   - Health on engine and component spares delivery to support the aftermarket
   - If applicable, a preliminary end-of-Production plan including a plan for manufacturing resources and asset/property transition
   - If applicable the cost - benefit analysis for ending production
6. An update to the Sustainment Plan (see section 6 of this document) highlighting
   - Updated sustainment strategy including product/services safety requirements plan, aftermarket support plan, and inventory strategy
   - A plan for repair inventory, as required/predicted and MRO strategy.
   - Completion of an estimate to the end of Sustainment Phase (to include customer/Market schedule and sustainment expectations, post-production aftermarket supportability plan)
7. An update to the defined 'Enterprise KPI' Deliverables (see Form 11721)
8. NRE Status on the program
9. A Global Trade review conducted and update the export / import plan, if required

10. A patent review (refer to EOP 258),
11. if applicable a communication plan including discontinuation notice for internal and external impacted parties.
12. A review of the sustainability and Environment, Health & Safety (EH&S) metrics for the complete product life cycle.
13. The intent of the above requirements shall be met during gate review using standard subgate E template (form 11760), using the gate requirement checklist (see section 11), open actions, major changes since last gated review and identified exit criteria actions

Gate E Authorization and Phase Closure

Gate E is called to review product performance and ensure lessons learned are captured in standard work for future programs.

5.3.8 GATE F – DISCONTINUATION OF SUPPORT

Gate F Trigger criteria

Gate F is required as needed to solicit agreement to discontinue support of a product or service

Gate Requirements

1. Status on Completion of contractual obligations and customer impacts review
2. Completion of discontinuation production strategy (to include plan for manufacturing resources, all government or customer furnished/owned property, the cost/benefit analysis for ending production, end-of-Production plan, including resource plan and asset/property transition)
3. "Completion of discontinuation sustainment strategy (to include safety requirements, aftermarket support plan, and inventory strategy, customer/Market schedule and sustainment expectations, post-production aftermarket supportability plan)"
4. A communication plan including discontinuation notice for internal and external impacted parties
5. The cost/benefit analysis for ending sustainment
6. Market analysis/Financial analysis that supports Program exit
7. Approved risk management plans, if necessary
8. Consensus on program exit strategy
9. Consensus on fleet retirement plan with level one schedule
10. Intellectual property provisions with approval by Legal
11. Consensus on trigger condition points through PMT disbanding

12. The intent of the above requirements shall be met during gate review using standard subgate F template , covering a requirement checklist , open actions, major changes since last gated review and identified exit criteria actions

6 MANAGEMENT PLANS

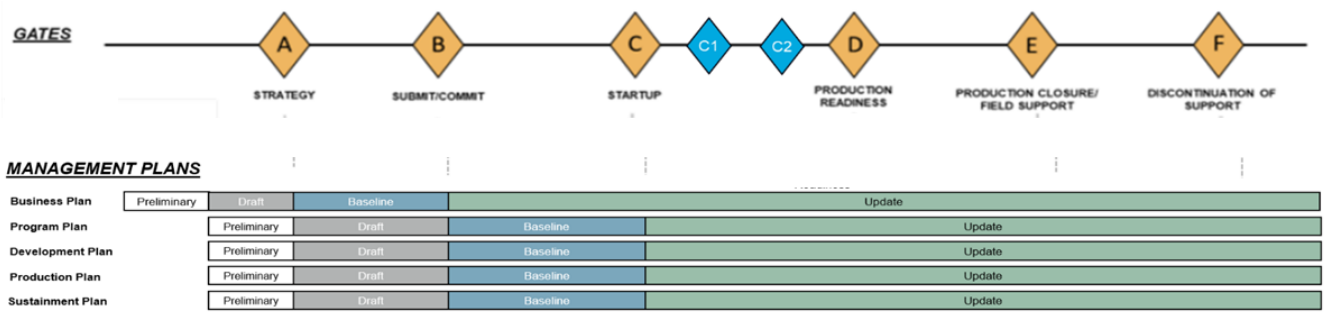
6.1 Purpose and Ownership of Management Plans

RLPM contains five Management Plans: Business, Program, Development, Production and Sustainment. Each Management Plan is owned by a specific entity and each Business/owner shall define the required contents for each management plan listed below to support major program gates and milestone reviews as such they will contain necessary information to support Gate Requirements highlighted in section 5.

All updates to plans will be reviewed and approved by the core PMMT members prior to major program gates.

The evolution of the maturity of the plans relative to the RLPM Gates is described in the Figure below.

- "Preliminary" plans contain a conceptual description of the activities to be performed, along with any associated assumptions. The plan contains enough detail to support the level of bidding and/or estimates required at the particular point in a program's life cycle. "Draft" plans add further details to the conceptual information and assumptions from the Preliminary Plan. The level of detail provided needs to be robust enough to support the level of bidding and commitment being made.
- "Baseline" plans contain detailed, executable activities that will meet the intended program requirements and that is ready for implementation.



6.2 Business Plan

The Business plan is owned by SFV Organization. The Business plan initially covers the market segment analysis, the opportunities with the proposed solutions and associated value propositions relative to key requirements as well a competitive assessment. This plan yields a marketing strategy and the associated financial details, required investments, all costs, and revenues of the Program. It matures from preliminary to draft before Gate A and is baselined by Gate B. At that point the Business plan is baselined and updated thereafter to track the evolution of the Program relative to the initial assumptions.

6.3 Program Plan

The Program plan is owned by the PMMT as per MMP 219. The Program plan captures the common set of program goals, specific KPI metrics and overall program strategy needed to meet all KPI. The Program plan includes a synopsis of the Program with a program overview, key members and stakeholders, a summary of the key contract deliverables to the customer and highlight of customer and regulatory requirements summarized in a PDRS requirements as per EOP 49. The program plan contains detail on the program resources, facilities and critical skills, and digital technology required to support program goals. The Program plan provides an overview of product architecture and keeps tracks of main risks and opportunities as well as the program execution strategy.

The program plan includes any tailoring to enterprise standard work (development, production, sustainment plans) required to support program objectives. It includes the master schedule (i.e. L1 plan) as well as the program specific criteria used to define and schedule gated reviews – where different to MMP220 guidance (see section 7.4 and section 5 of the current document).

The Program plan takes shape after Gate A and remains preliminary until Gate B. It is baselined and matures through Gate C after which it is updated and adjusted to reflect the activities required to meet the Program Business Plan. The elements of the Program plan and template are defined in EOP60.

6.3 Development Plan

The Development plan is owned by Engineering. The development plan contains the technical content planning (e.g. L2 and L3 plans), dependencies, development hardware procurement strategy, product maturity plans (e.g. TRL, MRL), test vehicle assets and main test configurations. It defines the formal technical reviews to review the requirements, such as the PDRS review, and to track the evolution of the design and validation activities such as sub-Gates C1 and C2. The Development plan takes shape after Gate A and remains preliminary until Gate B. It is baselined and matures through Gate C after which it is updated and adjusted to reflect the evolution of the design and validation activities and associated KPIs. The elements of the Development plan are defined in EOP 60.

6.4 Production Plan

The Production plan is owned by Operations. The Production plan ties into the Development plan for DFM, sourcing strategy (buy, make, build), and Partnership strategy as well as throughout the testing phase. It lays out the production capacities, locations and material flows; covers manufacturing approvals and all other industrialization aspects of the product. It ensures the flow down and implementation of the product definition, customer, quality, and regulatory requirements including an agreed producibility improvement & PPAP plan. The plan tracks and projects product costs and contains the specific production readiness dashboard (refer IMP 07-01-01) to be used at major program gates and milestones. The Production plan takes shape after Gate A and remains preliminary until Gate B. It is baselined and matures through Gate C after which it is updated and adjusted to reflect the evolution of the product needs considering obsolescence's, resourcing's and product improvements while the product remains in service. The elements of the Production plan are defined in the respected OPMP procedure.

6.5 Sustainment Plan

The Sustainment plan is owned by Customer Service. The Sustainment plan covers all aspect of a product while it is in service, from its introduction until it is discontinued. It covers meting the regulatory aspects (e.g. ICAs and SMS), contract and customer specific measures, aftermarket financials (including PPH contract management, warranty) , spare parts planning and provisioning, maintenance needs and strategies, as well as the service network (capability and capacity). The sustainment plan will contain key EIS readiness dashboard and standard In-service metrics to be used at major gates and program milestone review. The Sustainment plan takes shape after Gate A and remains preliminary until Gate B. It is baselined and matures through Gate C after which it is updated and

adjusted to reflect the evolution of the product needs throughout it's service life. The elements of the Sustainment plan are defined in CS-OP 41.

6.6 Tailoring Applicability

The intent of the tailoring process is to remove parts of the RLPM that are not necessary because that scope is not deemed applicable to the Program lifecycle.

The intent of tailoring is to adapt scope for project specifics and limited to what is applicable to the Program lifecycle, through the use of pre-defined archetypes (see section 7)

The tailoring includes management plan applicability. The Program Plans must include tailoring aligned with the archetype of the product and additional program specific tailoring using corresponding checklist form in section 11 of this document. See section 7 for details.

The Production, Sustainment and Development plans will use as required the program tailoring plan defined in the approved program plan.

7 PROGRAM ARCHETYPES & PROGRAM MANAGEMENT TAILORING

7.1 PURPOSE AND DEFINITION

The intent of the tailoring process is to remove parts of the RLPM that are not necessary because that scope is not deemed applicable to the Program lifecycle.

Business specific archetypes facilitate more efficient tailoring at the Program level. This includes management plan applicability and Entry/Exit criteria of each RLPM phase through aforementioned gated process.

Archetype is a defined grouping of like programs to accelerate tailoring for a type of contract applicability. Similar products or archetypes include similar quality requirements, engineering challenges, complexity, risk, applicable phase, large/complex/strategic, or contract type, enabling a more tailored and efficient program lifecycle management

7.2 P&WC ARCHETYPES

The following Archetypes are defined for P&WC programs

Ref	Description	Complexity	Gross Investment (USD)
1	Demonstrator	Minor	< 5M
2	Demonstrator	Minor	5-25 M
3	Demonstrator	Minor	25+ M
4	Demonstrator	Major	25+ M
5	Derivative	Minor	<5 M
6	Derivative	Minor	5-25 M
7	Derivative	Minor	25+ M
8	Derivative	Major	25+ M
9	Obsolescence/Resourcing/Field Upgrade	Minor	5-25M
10	New Offering	Major	25+ M

7.3 APPLICABILITY

The Management Plans will use the defined archetype of the product or service development. Details of this will be contained in the relevant BU operating procedures that contain management plan templates (ref EOP60, CSOP-41, and corresponding

7.4 RLPM GATE/REVIEW TAILORING

The following Tailoring applies to Lifecycle gates unless otherwise specified in approved PMMT program plan (see section 6.3).

Archetype Ref.	Description	Complexity	Total Program (USD)	Gate Tailoring					
				A	B	C	D	E	F
Archetype 1	Demonstrator	Minor	< 5M		✓	✓			
Archetype 2	Demonstrator	Minor	5-25 M		✓	✓			
Archetype 3	Demonstrator	Minor	25+ M		✓	✓			
Archetype 4	Demonstrator	Major	25+ M		✓	✓			
Archetype 5	Derivative	Minor	< 5M		✓	✓	✓	✓	✓
Archetype 6	Derivative	Minor	5-25 M		✓	✓	✓	✓	✓
Archetype 7	Derivative	Minor	25+ M	✓	✓	✓	✓	✓	✓
Archetype 8	Derivative	Major	25+ M	✓	✓	✓	✓	✓	✓

Archetype 9	Support/Product Upgrade	Minor	5-25 M			✓	✓	✓	✓
Archetype 10	New Offering	Major	25+ M	✓	✓	✓	✓	✓	✓

The approved Program plan as agreed by the PMMT, will define and substantiate the need for further specific program tailoring, including need of sub-gates.

Archetypes will also be used to tailor entry requirements to phases of the product lifecycle (see section 2.3 and 5.2 of the present document) using checklist in section 11 of this document for tailoring of requirements

Appendix B contains the tailoring of minimum entry requirements to program gates applicable for each Archetype.

8 METHOD OF APPLICATION

8.1 RLPM GATE REVIEWS

RLPM Gate reviews must be used in accordance with the hierarchy defined in section 7.4. As a minimum an Enterprise Technical (Fellows) Review, a Patent Review and a Business Review must be called prior to each RLPM Gate Review or otherwise identified in the gate entry/exit requirements. The intent of these reviews is to identify main program risks (EOP 168) and to align for PMMT to have management plans alignment prior to lifecycle phase reviews.

Final content of RLPM Gate review presentation must be reviewed to an Executive level within each business unit prior to the gate review.

8.1.1 The PMMT must appoint a RLPM Gate Lead (refer to Appendix A), to determine when phase entry/exit permission is sought based on program readiness for the gate per the definitions above.

8.1.2 All RLPM Gate reviews must be scheduled by the Quality department with input from the PMMT, RLPM Gate Lead and Executive calendar.

8.1.3 The RLPM Gate Lead must schedule a preparation kick-off meeting prior to the Gate Review.

8.1.4 The RLPM Gate Lead (refer to Appendix A) must organize an Enterprise Technical Review, by contacting the Chief Engineer's Office representative to schedule a Fellow's review including risk assessment.

8.1.5 The RLPM Gate Lead (refer to Appendix A) must organize a Business Review, by contacting the Marketing and Strategic Financial Valuation (SFV) representatives to schedule a Business Review.

8.1.6 The RLPM Gate Lead must ensure that Management plans are approved by PMMT prior to Gate reviews.

8.1.7 Only approved templates are to be used.

Templates for RLPM Gate Reviews are available in Forms Control (refer to section 11 of current document for form numbers). Templates are divided into 2 layers:

Mandatory Presentation - Slides are provided in the required format to cover all gate requirements. Additional supporting slides can be added to detail the mitigation for red risks.

Additional Content - Any additional slides that serve as a record of the program status.

8.1.8 The status of actions identified at previous RLPM Gate reviews must also be reviewed during preparation.

8.2 P&W PRESIDENT RLPM GATE REVIEW (PW PRES)

When required, a Presidential RLPM Gate (PWPRES) must be conducted as per WP-1.5.1 following the internal gate review per this procedure.

PW PRES level gated reviews are required according to the following investment thresholds:

- For Gate A & Gate B where E&D ≥ 25 M$
- For Gates C to Gate F where E&D ≥ 100 M$
- For Gate A to Gate F where External funding ≥ 500M$

PW PRES level reviews are required for RLPM gates only and not for P&WC sub-gates or milestones.

To schedule a PW PRES RLPM Gate Review, a P&WC EXEC accountable for a given RLPM Gate (or a delegate) should send a request to the administrative assistant of PW VP of Quality with the following information, at least 1 month prior to desired date of the review:

Purpose	A clearly written statement of the meeting goal. Describes the type of meeting and should be simple and focused. (i.e., Decision meeting/flow down/brainstorming)
People	Who needs to be in the meeting. (i.e., Critical participants only, in additional to standard quorum)
Preview	Advanced view of the discussion topics/agenda (if avail)
Process	Describe the type of meeting. (i.e., formal/informal, slides / no slides, video/no video)

8.3 DELTA GATES

Delta gates, where appropriate, are convened to review program progress when:

- A boundary condition agreed to at a prior program Lifecycle Phase review or a job ticket requirement is exceeded,
- A new risk or capability / capacity shortcoming emerge that put significant Job Ticket objectives at risk,
- The planned investment estimate at completion exceeds the approved AR investment,
- Key business or technical decisions must be assessed between formal Lifecycle Phase review, or
- A significant change in the business context or scope

8.4 RLPM GATE REVIEW WAIVER

8.4.1 A request for authorization to waive an RLPM Gate review must be submitted to Senior Management by the RLPM Gate Lead (refer to Form 11723). The following up to date documents must be included with the request:

- The 'Enterprise Key Performance Indicators' (refer to Form 11721),
- Status of the Production Readiness Dashboard Exit Criteria (refer to IMP 07-01-01) for the Lifecycle Phase review to be waived.
- Export licensing status given the sourcing plan and scope changes
- The Business Case,
- The Risk Cube,
- Any Export Control / CGP compliance plan, and
- Product IP Strategy Plan (refer to EOP 258).

8.4.2 The Waiver must be approved by the PWC President. Once approved the Waiver is stored in the appropriate RLPM Gate Repository folder.

8.5 CONDUCT RLPM GATE REVIEWS

8.5.1 The RLPM Gate Review Lead must open the meeting with a summary of the meeting protocol and the assignment of roles during the meeting and then facilitates the review.

Note: Where applicable, a list of key questions to be addressed is provided in the RLPM Gate review template. This list is to be used as a guide to presenters for mandatory and supporting material to be presented. In addition, the same list may be used by reviewers as key questions to ask. In getting to a RLPM Gate review, technical questions should be addressed during the technical review, while business questions should be covered during the business review.

8.5.2 RLPM Gate Review Board (RLPM-GRB) Members - Quorum is defined by the membership. At the time of the RLPM Gate review the President's Office may allow the RLPM Gate review to proceed with a reduced attendance.

Qty.	Position	Alternate
1[1,]	President	Delegated VP
1[2,]		Operations Director WSK
1	VP Engineering	VP Engine Development Programs
1[2,]	Head of Design Organization (Engineering Director), WSK	VP Engineering, P&WC
1	VP Marketing or VP Business Development	Director Marketing or Director Business Development
1	VP Legal	Director Legal
1	VP Customer Service	Director Customer Support and / or Director Service Center
1	VP Operations	VP Supply Management or Director of Operations or Director Industrialization
1	VP Finance	Senior Manager - Financial Planning & Analysis, Controller or Engine Business Finance Manager Controller, or Strategic Financial Valuation Dept Manager
1	VP Quality	Director / Fellow Quality
1	Project Director or Program Manager	n/a

1	Chief Engineer, Product Integrity	Delegate
1	VP Industrialization	Delegate

[1] Optional for Gate A
[2] For EASA APU program RLPM Gate Reviews

8.5.3 To allow for RLPM Gate Review meeting effectiveness and constructive discussion, attendees should be limited as much as possible to the Quorum defined above.

8.5.4 Refer to WP 1.5.1 (PW document) for PWPRES level RLPM-GRB quorum requirements.

8.5.5 Business case elements such as cost may require a review with reduced attendance.

8.5.6 A list of attendees must be kept by RLPM Gate Lead.

8.5.7 The RLPM Gate Lead must arrange for meeting minutes and action items to be taken during the RLPM Gate Review using the RLPM Gate Review Process (form 11757).

8.6 ISSUE MINUTES / ACTIONS

8.6.1 The RLPM Gate Lead must ensure that the President signs a copy of the RLPM Gate Review Closure Report (refer to Form 11299). If the President does not attend a RLPM Gate Review then the Alternate listed in Table 8.5.2 must sign as delegate. The RLPM Gate Review presentation and the RLPM Gate Review (form 11757) must be attached to the closure report.

8.6.2 All final RLPM Gate Review presentations, RLPM Gate Review closure report (form 11299) and actions (form 11757) must be filed to the official RLPM Gate Review Repository site under the corresponding program by the Quality Representative, with a notification to the PMMT, PMT & actioners.

8.6.3 The RLPM Gate Lead must ensure that RLPM office representatives signs form 11757 as an acknowledgement that all applicable documents and approvals are filed to the official RLPM Repository site including signed form 11299, RLPM Gate Presentation Material and completed form 11757.

8.6.4 Progress on the closure of action items must be reported in the forums used by the PMMT.

8.6.5 Action item responses must be reviewed by the PMMT / PMT and approved.

9 LIFECYCLE PHASE REVIEW (LPR) DISPOSITION

9.1 GO - Full Approval

Authorization to proceed to the next Program Phase.

9.2 Conditional Approval

A Conditional RLPM Gate approval is granted to allow program continuance while open issues are worked to closure. All issues are closed via the action item list issued following the RLPM Gate. A program review meeting or Delta Gate must be scheduled when the conditional approval is granted.

9.3 REDIRECT

An RLPM gate redirect is granted when RLPM-GRB establishes a change in scope or determines information is incomplete. A follow up program review meeting or Delta Gate must be scheduled following a REDIRECT Gate outcome to ask permission to continue to next program gate.

A new RLPM Gate Review Closure Report must be signed by the Office of the President and attached to the REDIRECT approval as an official confirmation that the RLPM Gate review has achieved a Full or Conditional Approval following action closure.

9.4 NO GO - Rejected

An RLPM Gate review is rejected when the technical proposal or business case are not acceptable. All program activities must be suspended until the Gate is approved

9.4.1 In order for the program to proceed to the next phase, the PMMT must schedule a repetition of the current RLPM Gate review.

10 RESPONSIBILITIES

10.1 Based on the program needs, the agenda for the RLPM Gate Review will be agreed between the RLPM Gate process owner and the PMMT Lead for the applicable gate using the agreed Program plan tailoring as required.

10.2 The Marketing Account Manager is responsible for:

- Issuing a request for creation of a program file structure in the RLPM Repository, to the Quality Representative.

10.3 The Vice-President, Quality is responsible to appoint a Quality Representative who is responsible for:

- Maintaining the system and process for program RLPM reviews,
- Establishing the executive level program RLPM review schedule in conjunction with the RLPM-GRB.
- Creating the file structure for the new Program within the RLPM Repository site,
- Ensuring that a 'RLPM Gate Closure Report' has been added to the repository following a Gate review.

10.4 The RLPM-GRB (RLPM-Gate Review Board) as defined in section 8 "Conduct RLPM Gate Reviews" of this document is responsible for the success of the program and for providing program resources. Their responsibilities include:

- Approving any major changes to the program plan / level 1 schedule milestones and resource requirements,
- Ensuring changes are reflected in an updated business case,
- Ensuring the proposed program is still in line with the Company business strategy and AR
- Ensuring all identified key issues are adequately addressed and key issues are not missed or overlooked,
- Assessing the adequacy of the data presented to justify the program assumptions decisions or proposals,
- Raising and discussing issues collectively that are pertinent to deciding if the program should proceed, and
- Accepting the boundary conditions / trigger points proposed by the PMMT lead or modifying them with the program concurrence.

10.5 The RLPM Gate Lead is responsible for the successful outcome of each Gate review. Responsibilities include:

- Ensuring appropriate functional reviews are held,
- Ensuring the program RLPM Gate covers the proper content,
- Ensuring that the appropriate knowledgeable people attend each RLPM Gate review,
- Ensuring that data is presented at the Gated reviews to validate decisions,
- Ensuring that action items are documented & assigned to minimize program risks,
- Presenting recommended trigger points and boundary conditions at each program RLPM Gate review,
- Scheduling trigger reviews when a boundary condition is exceeded, and to endorse proposed contingency options for identified trigger point risk items (e.g. Fellows' review),
- Ensuring all likely sources of risk have been identified and quantified,
- Ensuring the proposed risk mitigation plan is feasible and the risk accepted by the program is manageable,
- Ensuring the content covered during the Gate review is complete, relevant to the phase or trigger issue and all program assumptions decisions or proposals are supported with data,
- Ensuring that the agreement resulting from the Gate review is clearly specified (i.e. decision / agreement reached, including trigger points, boundary conditions, schedule / budget / allocation and next Gate, etc.),
- Upload the RLPM Gate Review Closure report to the appropriate RLPM repository for record retention, and
- Review and follow-up of the previous RLPM Gate open actions prior to next RLPM Gate Review.

10.6 The **Accountable Program Executive** (responsible for the successful outcome of each RLPM Gate review) responsibilities include:

- PW PRES RLPM Gate Review process requirements are reviewed and followed (Ref WP1.5.1),
- PW PRES RLPM Gate Review is scheduled as required (Ref Para 8.2),
- Quorum is verified and recorded,
- Objective and recommended trigger points are clearly stated,
- Attendance and actions are documented,
- Ensuring that the agreement resulting from the RLPM Gate review is reached,
- RLPM Gate Review Closure Report form 9348 or equivalent document (containing as a minimum the requirements in form 9348) is generated,
- Ensure copy of the presentation and complete RLPM Gate Review Closure Report is provided to P&WC RLPM Office Representative (Quality Executive

Assistant) to be uploaded in the PW RLPM Records Repository within 30 days of completion of the RLPM Gate Review

11 REFERENCES

11.1 The following documents form a part of this procedure to the extent specified herein.

11.2 MANAGEMENT MANUAL PROCEDURES

MMP 60	Protection and Control of Intellectual Property
MMP 150	Contracts
MMP 219	Integrated Product Management Process
MMP 375	International Trade Compliance
MMP 444	Marketing Proposal Inquiries
MMP 501	New Program Introduction – FSC Management
MMP 515	Engine Program Investment Guidelines (CAR's)

11.3 ENGINEERING OPERATING PROCEDURES

EOP 49	Product Design Requirements Specification
EOP 60	Management of Engine Development Programs
EOP 103	Design Process and Design Reviews
EOP 153	Design for Sustainability, Environment, Health & Safety
EOP 158	Engine Development / Production Discrepancy System
EOP 168	Project Risk Management Process
EOP 174	Scope Management
EOP 258	Patent Risks and Responsibilities

11.4 OPERATIONS PROCEDURES

IMP 07-01-01	Production Readiness Process

11.5 CUSTOMER SUPPORT OPERATING PROCEDURES

CSOP P-41	Customer Service Field Readiness (EIS) Process

11.6 QUALITY MANAGEMENT PROCEDURES & STANDARDS

QMSP 07-05-02	PPAP
QMSP 10-06	Manufacturing Approval – Engines & Sub-Assemblies
ASQR-01	Supplier Quality System Requirements
AS 9145	SAE APQP Standard

11.7 PRATT & WHITNEY WORLDWIDE PROCESS

WP 1.5.1	WP 1.5 RTX Lifecycle Program Management

11.8 LRP PROCESS MAPS

PDD0250	Design Review Process

11.9 FORMS

11299	RLPM Closure Report Closure Report
11720	Job Ticket
11721	Enterprise Key Performance Indicators
11723	Authorization to Waive RLPM Gate
11757	RLPM Process Review
	Program Tailoring checklist

11.10 Templates

11758	Gate A	Presentation
11759	Gate B	Presentation
11760	Gate C	Presentation
	Sub Gate C.1	Presentation
11761	Sub-Gate C2	Presentation
11762	Gate D	Presentation
11763	Gate E	Presentation
	Gate F	Presentation

12 FREQUENCY OF REVIEW

12.1 This Procedure shall be reviewed every two years by the Monitor. A revision log is maintained in order to keep track of all the changes. When an entity would like to perform a change before the review every two years, they must submit the proposed revision to the Monitor and the Management Controls Group.

PRATT & WHITNEY CANADA CORP.

RTX LIFECYCLE PROGRAM MANAGEMENT (RLPM) REVIEW PROCESS – MMP 220

REVISION STATUS

REVISION	Review Date (MM-YY)	DESCRIPTION
0		
1		
2		
3		
4		
5		
6		
7		
8		
9		
10	07-2015	Rewrite, Addition of APQP, alignment of process
11	02-2018	Addition of sustainability & Eco-Design, change to record retention policy
12	11-2020	Alignment to Production Readiness, APQP and further refinement.
13	04-2025	Alignment to P&W WP 1.5 (RLPM), addition of Accountable Executive
14		
15		

6 APPENDICES

APPENDIX A - Lead by Phase and Gate:

Gate/ Sub-Gate	Program Phase	Accountable Executive	RLPM Lead
A	Strategy Gate	Strategic Planning Executive	Strategic Planning Manager
B	Submit/ Commit Gate	Marketing or BD Program Executive	Marketing Account Manager
C, C1 and C2	Start-up & Detailed Design Sub-Gates	Engineering Program Executive	Engineering Program Manager
D	Production Readiness Gate	Operations Executive	Operations Program Manager
E	Field Support & Production Closure Gate	Customer Service Executive	Customer Service Manager
F	Discontinuation of Support Gate	Engineering Program Executive	Engineering Project Manager

APPENDIX A – Gate Requirements Tailoring per Archetype

Gate A - Program Strategy Gate

	Archetype Reference	1	2	3	4	5	6	7	8	9	10*
1	Pursue/No pursue decision							✓	✓		✓
2	Program Management Team (PMT)/Capture team assigned							✓	✓		✓
3	Baseline capture plan as part of the Program Plan							✓	✓		✓
4	Establish requirement for partners participation							✓	✓		✓
5	Draft Business Plan, as per section 6 of this document, highlighting								✓		✓
5.1	Marketing Segmentation and Environment							✓	✓		✓
5.2	Opportunity Analysis completed							✓	✓		✓
5.3	Marketing Strategy reviewed							✓	✓		✓
6	Preliminary Development, Production, and Sustainment Plans as highlighted in Section 6 of this document								✓		✓
7	A Factory Standard Cost (FSC) study (refer to MMP 501) summarized in preliminary Production plan							✓	✓		✓
8	A Preliminary Program plan as highlighted in section 6 of this document, containing								✓		✓
8.1	A review of Technical Configuration							✓	✓		✓
8.2	A comprehensive assessment of the overall program risk associated with the enterprise program plan							✓	✓		✓
8.3	Phase funding, schedule and resources/investments authorized							✓	✓		✓
9	A product IP strategy plan including patent review schedule (refer to EOP 258)							✓	✓		✓
10	The 'Enterprise KPI' Deliverable targets (refer to Form 11721)							✓	✓		✓
11	A Global Trade review conducted and baseline Export/Import (EXIM) Plan, if required							✓	✓		✓
12	IRs completed per plan								✓		✓
13	The sustainable engine strategy to meet company goals (refer to EOP 153)								✓		✓
14	Tailoring review/approval							✓	✓		✓

* Archetype 10 is aligned with the full entry and exit criteria defined in section 5.2 of this document

Gate B - Proposal Submittal gate

Archetype Reference	1	2	3	4	5	6	7	8	9	10*
1 A baseline Business plan (see section 6 of this document) highlighting	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
1.1 Market Summary including a competitive Analysis, Product positioning, Market outlook-volume	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
1.2 A summary of the business case (refer to MMP515) and contract deliverables	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
1.3 Required phase funding, schedule and resources/investments	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
1.4 Management Reserve identified per P&W WP 1.5 aligned with program risks identified (refer to MMP515 for standard work)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2 A Draft program Plan (see section 6 of this document) highlighting	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2.2 Funded TRL/MRL maturation plan aligned with requirements in EOP 31 – Engineering Cost estimates (refer to MMP515 for standard work)						✓	✓	✓	✓	✓
2.3 Labour requirements from all Business units to support program L1 plan, associated risks and critical skills identified.	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2.4 Level 1 Plan (Creation of the proposed timeline including AR approval)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2.5 The intended partners and/or suppliers and the proposed quality plans as well as any additional company requirements	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2.6 The program archetype and tailoring review/approval for this program	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2.7 A Partnership/Collaboration strategy, including Partnership assessment (per MMP 507), workscope distribution (per EOP 102), and Program share allocation						✓	✓	✓		✓
3 A Commercial Response to Customer - OEM request overview, Commercial Opportunity Strategic Sandbox (COSS) and Commercial Response to Customer (Proposal Sandbox)						✓	✓	✓		✓
4 A summary of the technical proposal that was held prior to the Passport review (refer to EOP 60, 103 & PDD0250)		✓	✓	✓		✓	✓	✓	✓	✓
5 A comprehensive assessment of the overall program risk associated with the enterprise program plan, including:		✓	✓	✓	✓	✓	✓	✓	✓	✓
5.1 All items identified as Red must have a mitigation plan presented (Including status of required resources financial and labour)		✓	✓	✓	✓	✓	✓	✓	✓	✓
6 A Draft Production Plan (see section 6 of this document)					✓	✓	✓	✓	✓	✓
6.1 A review of the Factory Standard Cost (FSC)					✓	✓	✓	✓	✓	✓
6.2 A preliminary Industrialisation plan					✓	✓	✓	✓	✓	✓
6.3 Production Readiness Dashboard Exit Criteria (refer to IMP 07-01-01)					✓	✓	✓	✓		✓
7 The 'Enterprise KPI' Deliverables (refer to Form 11721)					✓	✓	✓	✓	✓	✓
8 A Draft Sustainement plan (see section 6 of this document) highlighting the overall aftermarket strategy					✓	✓	✓	✓	✓	✓
9 A patent review (refer to EOP 258)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
10 Global Trade (GT) assessment status and export licensing plan (Birth Certificate) (2)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
11 Draft Development Plan (see section 6 of this document)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
12 A program communication strategy					✓	✓	✓	✓		✓
13 The program sustainability requirements for the complete product life cycle to meet company and customer goals					✓	✓	✓	✓		✓
14 The intent of the above requirements shall be met during gate review using standard Gate B template (form 11760), using the gate requirement checklist (see section 11), open actions, major changes since last gated review and identified exit criteria actions	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

* Archetype 10 is aligned with the full entry and exit criteria's defined in section 5.2 of this document

Gate C - Start-up Gate

	Archetype Reference	1	2	3	4	5	6	7	8	9	10
1	A baselined program Plan (see section 6 of this document) highlighting	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
1.1	An agreed set of program goals	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
1.2	An approved Program resources ; finances, people and critical skills, facilities/capital investment, and digital technology to launch full development	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
1.3	A status from Business units on labour requirements identified in Gate B to support program L1 plan, associated risks and mitigation plans	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
1.4	An agreed L1 Plan	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
1.5	A Product overview and an agreed architecture concept	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
1.6	A PMMT agreed PDRS (see EOP 49) containing regulatory requirements, key customer requirements and Internal requirements						✓	✓	✓	✓	✓
1.7	Any identified non-compliances to PDRS necessary to meet program goals, agreed by PMMT and summarized in program plan						✓	✓	✓	✓	✓
2	Independent reviews completed per program plan		✓	✓	✓	✓	✓	✓	✓	✓	✓
3	An approved capital investment plan and NRE status	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
4	An updated Business plan (see section 6 of this document) highlighting	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
4.1	A Marketing outlook summary			✓	✓	✓	✓	✓	✓	✓	✓
4.2	A review of the business case summary (refer to MMP515) and update to Management reserve (as per IR review)					✓	✓	✓	✓	✓	✓
4.3	Key contract deliverables	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
4.4	Any change in scope impacting the AR or the business case	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
5	A baselined Production Plan (see section 6 of this document) highlighting	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
5.1	An initial assessment of facility, site, and infrastructure plans and requirements.					✓	✓	✓	✓	✓	✓
5.2	An internal initial production capacity plan and/or factory model with a plan to review maturity during technical reviews, including manufacturing automation plans.					✓	✓	✓	✓	✓	✓
5.3	Sourcing and Manufacturing Strategy (including PPAP plan with a list of unique part numbers including mitigation for common hardware issues) (refer to QMSP 07-05-02),	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
5.4	An initial Production Readiness Dashboard Exit Criteria (refer to IMP 07-01-01)					✓	✓	✓	✓	✓	✓
5.5	A Factory Standard Cost (FSC) summary and plan					✓	✓	✓	✓	✓	✓
6	A Comprehensive assessment of the overall program risk associated with the enterprise program plan, including:					✓	✓	✓	✓	✓	✓
6.1	A Summary of the technical risk identified at the technical review that was held prior to the Passport review (refer to EOP 103 & PDD0250),		✓	✓	✓		✓	✓	✓	✓	✓
6.2	All items identified as Red must have a mitigation plan presented (Including status of required resources financial and labour)		✓	✓	✓		✓	✓	✓	✓	✓
7	TRL and MRL 6, whichever is greater. If this is not achievable					✓	✓	✓	✓	✓	✓
7.1	There must be a proposed TRL/MRL maturation plan to result in achieving TRL level 6 by Preliminary Design Review (PDR) and an MRL 6 by Critical Design Review (CDR).					✓	✓	✓	✓	✓	✓
7.2	Funding of the TRL and MRL maturation plan must occur and include back up plans if unable to achieve the proposed TRL/MRL 6 level. TRL/MRL maturation plan shall be aligned with requirements in EOP 31 – Engineering Cost estimates and MMP515 guidance						✓	✓	✓	✓	✓
7.3	The back up plans must meet both the Program and Business objectives.					✓	✓	✓	✓	✓	✓
8	A baselined Development Plan (see section 6 of this document) that defines the technical execution strategy	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
8.1	A summary of the preliminary Verification and Validation (V&V) plan highlighting key methods identified assocaited with major investments (ref. EOP 60 for process and validation methodologies)						✓	✓	✓	✓	✓
8.2	As a function of partnership risk profile, and assigned scope, a dedicated partnership development plan may be required						✓	✓	✓	✓	✓
9	Global ITC assessment status and export licensing given the sourcing plan and scope changes					✓	✓	✓	✓	✓	✓
10	A patent review (refer to EOP 258),				✓	✓	✓	✓	✓	✓	✓
11	A baselined Sustainement Plan (see section 6 of this document) that highlights the overall Aftermarket strategy					✓	✓	✓	✓	✓	✓
12	A review of the sustainability and Environment, Health & Safety (EH&S) metrics for the complete product life cycle						✓	✓	✓	✓	✓
13	The 'Enterprise KPI' Deliverables (see Form 11721),					✓	✓	✓	✓	✓	✓
14	The intent of the above requirements shall be met during gate review using standard Gate C template (form 11760), using the gate requirement checklist (see section 11), open actions, major changes since last gated review and identified exit criteria actions	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

* Archetype 10 is aligned with the full entry and exit criteria defined in section 5.2 of this document

Gate D - Production Readiness Gate

	Archetype Reference	1	2	3	4	5	6	7	8	9	10*
1	An updated program Plan (see section 6 of this document) highlighting					✓	✓	✓	✓	✓	✓
1.1	An update on the program status (strategy, marketing, economic & aftermarket)					✓	✓	✓	✓	✓	✓
1.2	Approved Program resources (finances, people, facilities/capital investment, and digital technology) to launch serialized production					✓	✓	✓	✓	✓	✓
1.3	An update of the L1 Plan					✓	✓	✓	✓	✓	✓
1.4	The 'enterprise KPI' Deliverables (see form 11721)					✓	✓	✓	✓	✓	✓
1.5	Successful completion of or plan for completion of V&V of component, product, and system level requirements with customer approval documented.					✓	✓	✓	✓	✓	✓
1.6	A compliance status against the PDRS (refer to EOP 49) where required with a time-bound, resourced and funded closure plan or associated risks reflected in enterprise program risk cube						✓	✓	✓	✓	✓
1.7	Identification of funded program improvement plans based on customer/airframe Market Feedback Analysis, a Customer Oriented Results & Excellence (CORE) method, as required						✓	✓	✓	✓	✓
1.8	A Product performance and architecture stability overview (including proof of manufacturing with lessons learned incorporated into design or manufacturing documentation)					✓	✓	✓	✓	✓	✓
1.9	The release both the Technical Data Package (TDP - including definition, ESA and SFI approvals) and Manufacturing Data Package (MDP - including MOS, AFS and approved repairs).					✓	✓	✓	✓	✓	✓
1.1	Conversion of the engineering BOM (eBOM) into manufacturing BOM (mBOM).					✓	✓	✓	✓	✓	✓
2	An updated development plan (see section 6 of this document) highlighting					✓	✓	✓	✓	✓	✓
2.1	Completion of all qualification testing and completion of all certification reports with remaining open items identified					✓	✓	✓	✓	✓	✓
2.2	Completion of all verificaiton activities for first production hardware (reference EOP 57) including PFMEA, DFMEA, FAIR.					✓	✓	✓	✓	✓	✓
2.3	Status of compliance against certification requirements					✓	✓	✓	✓	✓	✓
3	An updated Business plan (see section 6 of this document) highlighting					✓	✓	✓	✓	✓	✓
3.1	An update marketing outlook summary					✓	✓	✓	✓	✓	✓
3.2	A review of the business case summary (refer to MMP515) and required update to Management reserve (as per IR review)					✓	✓	✓	✓		✓
3.3	An updated status on key contract deliverables					✓	✓	✓	✓	✓	✓
3.4	Any change in scope impacting the AR or the business case					✓	✓	✓	✓	✓	✓
4	An update to the Production Plan (see section 6 of this document) highlighting					✓	✓	✓	✓	✓	✓
4.1	Released production BOM and production sources identified for all parts with approved rate plans.					✓	✓	✓	✓	✓	✓
4.2	Gap assessment results of facility, site, and infrastructure plans and requirements in support of serialised production						✓	✓	✓	✓	✓
4.3	A Factory Standard Cost (FSC) summary and plan established and funded					✓	✓	✓	✓	✓	✓
4.4	An updated Production Readiness Dashboard Exit Criteria (refer to IMP 07-01-01)					✓	✓	✓	✓	✓	✓
4.5	Industrial Strategy (including PPAP plan status as per QMSP 07-05-02) with resources in place (to include but not limited to staffing, equipment, tooling, and fixtures)						✓	✓	✓		✓
4.6	Documentation of Industrial risks and technical debt at Gate D with funded mitigation plans in place and communication to the customer as required					✓	✓	✓	✓	✓	✓
4.7	A list of identified post certification changes at gate D that require customer or field retrofit (OEUP) and consequential financial impact to business plan assumptions						✓	✓	✓		✓
5	Verification that all DR's having Hazardous Engine Condition or EH&S implications are closed with corrective action (refer to EOP 158), and evidence that all other DR have action plans					✓	✓	✓	✓	✓	✓
6	A Comprehensive assessment of the overall program risk associated with the enterprise program plan, including:					✓	✓	✓	✓	✓	✓
6.1	A summary of the Industrialization and technical risk identified at the technical review that was held prior to the Gated review (refer to EOP 103 & PDD0250),						✓	✓	✓	✓	✓
6.2	All items identified as Red must have a mitigation plan presented (Including status of required resources financial and labour)						✓	✓	✓	✓	✓
6.3	Identified risks requiring fleet data input (EIS Excellence plan - including MRO or field inspections or DPHM data to support ICA completion)					✓	✓	✓	✓	✓	✓
7	TRL 8 and MRL 9, whichever is greater. If this is not achievable					✓	✓	✓	✓	✓	✓
7.1	There must be a proposed TRL/MRL maturation plan to result in achieving TRL level 8 and MRL 9 by Entry into Production (EIP).					✓	✓	✓	✓	✓	✓
7.2	Funding and resource allocation plan of the TRL and MRL maturation plan must occur and include back up plans if unable to achieve the proposed TRL 8 and MRL 9 level.					✓	✓	✓	✓	✓	✓
7.3	The back up plans must meet both the Program and Business objectives.					✓	✓	✓	✓	✓	✓
8	Global Trade assessment status and export licensing plan (Birth Certificate) given the sourcing plan and scope changes						✓	✓	✓		✓
9	A patent review (refer to EOP 258), along with documentation Legal/Contracts/Compliance of program developed IP including sustainment strategies						✓	✓	✓		✓
10	NRE status					✓	✓	✓	✓		✓
11	An update to Sustainement Plan (see section 6 of this document) highlighting the following					✓	✓	✓	✓		✓
11.1	Completion of the sustainment customer support strategy plan (covering equipment, manuals, repair strategy, and training)					✓	✓	✓	✓	✓	✓
13	A review of the sustainability and Environment, Health & Safety (EH&S) metrics for the complete product life cycle						✓	✓	✓	✓	✓
14	The intent of the above requirements shall be met during gate review using standard subgate D template (form 11760), using the gate requirement checklist (see section 11), open actions, major changes since last gated review and identified exit criteria actions					✓	✓	✓	✓	✓	✓

* Archetype 10 is aligned with the full entry and exit criteria defined in section 5.2 of this document

Gate E - Production Closure/Field Support Gate

	Archetype Reference	1	2	3	4	5	6	7	8	9	10
1	An updated program Plan (see section 6 of this document) highlighting					✓	✓	✓	✓	✓	✓
1.2	A compliance to program goals including customer and internal requirements (identified gaps to PDRS)					✓	✓	✓	✓	✓	✓
1.2	An update of the L1 Plan					✓	✓	✓	✓	✓	✓
1.3	Identification of funded program improvement plans based on customer/airframe Market Feedback Analysis, a Customer Oriented Results & Excellence (CORE) method, as required					✓	✓	✓	✓	✓	✓
2	An updated Business Plan (see section 6 of this document) highlighting						✓	✓	✓	✓	✓
2.1	A review of the business case summary (refer to MMP515) and required update to Management reserve (as per IR review)						✓	✓	✓	✓	✓
2.2	Performance against contractual obligation assessment to determine customer and business case impact					✓	✓	✓	✓	✓	✓
3	A Comprehensive assessment of the overall program risk associated with the enterprise program plan, including:					✓	✓	✓	✓	✓	✓
3.1	A Summary of the technical risk identified at the technical review that was held prior to the Passport review (refer to EOP 103 & PDD0250),						✓	✓	✓	✓	✓
3.2	All items identified as Red must have a mitigation plan presented and associated post certificaton investment financial resources (technical debt) required (aligned with AR)						✓	✓	✓	✓	✓
3.3	Identified risks requiring fleet data input (EIS Excellence plan - including MRO or field inspections or DPHM data to support ICA completion)					✓	✓	✓	✓	✓	✓
4	Capture and communicate enterprise lessons learned for future programs					✓	✓	✓	✓	✓	✓
5	An update to the Production Plan (see section 6 of this document) highlighting					✓	✓	✓	✓	✓	✓
6	An update to the manufacturing resources (such as staff and equipment) plan					✓	✓	✓	✓	✓	✓
6.1	An update on Factory standard Cost					✓	✓	✓	✓	✓	✓
6.2	If applicable a plan for all government or customer furnished/owned property					✓	✓	✓	✓	✓	✓
6.3	If applicable, completion of all production contractual requirements					✓	✓	✓	✓	✓	✓
6.4	Industrialization Status, Production Readiness Dashboard Exit Criteria (refer to IMP 07-01-01)					✓	✓	✓	✓	✓	✓
6.5	Completion of MRL assessments (demonstrate that product is MRL9, show MRL readiness plan for new product changes)					✓	✓	✓	✓	✓	✓
6.6	Health on engine and component spares delivery to support the aftermarket					✓	✓	✓	✓	✓	✓
6.7	If applicable, a preliminary end-of-Production plan including a plan for manufacturing resources and asset/property transition					✓	✓	✓	✓	✓	✓
6.8	If applicable the cost - benefit analysis for ending production					✓	✓	✓	✓	✓	✓
7	An update to the Sustainement Plan (see section 6 of this document) highlighting					✓	✓	✓	✓	✓	✓
7.1	Updated sustainment strategy including product/services safety requirements plan, aftermarket support plan, and inventory strategy					✓	✓	✓	✓	✓	✓
7.2	A plan for repair inventory, as required/predicted and MRO strategy.						✓	✓	✓	✓	✓
7.3	Completion of an estimate to the end of Sustainment Phase (to include customer/Market schedule and sustainment expectations, post-production aftermarket supportability plan)					✓	✓	✓	✓	✓	✓
8	An update to the defined 'Enterprise KPI' Deliverables (see Form 11721)					✓	✓	✓	✓	✓	✓
9	NRE Status on the program							✓	✓	✓	✓
10	A Global Trade review conducted and update the export / import plan, if required					✓	✓	✓	✓	✓	✓
11	A patent review (refer to EOP 258),						✓	✓	✓	✓	✓
12	if applicable a communication plan including discontinuation notice for internal and external impacted parties.					✓	✓	✓	✓	✓	✓
13	A review of the sustainability and Environment, Health & Safety (EH&S) metrics for the complete product life cycle.						✓	✓	✓	✓	✓
14	The intent of the above requirements shall be met during gate review using standard subgate E template (form 11760), using the gate requirement checklist (see section 11), open actions, major changes since last gated review and identified exit criteria actions					✓	✓	✓	✓	✓	✓

* Archetype 10 is aligned with the full entry and exit criteria defined in section 5.2 of this document

Gate F - Discontinuation of support Gate

	Archetype Reference	1	2	3	4	5	6	7	8	9	10*
1	Status on Completion of contractual obligations and customer impacts review					✓	✓	✓	✓		✓
2	Completion of discontinuation production strategy (to include plan for manufacturing resources, all government or customer furnished/owned property, the cost/benefit analysis for ending production, end-of-Production plan, including resource plan and asset/property transition)					✓	✓	✓	✓		✓
3	Completion of discontinuation sustainment strategy (to include safety requirements, aftermarket support plan, and inventory strategy, customer/Market schedule and sustainment expectations, post-production aftermarket supportability plan)					✓	✓	✓	✓		✓
4	A communication plan including discontinuation notice for internal and external impacted parties					✓	✓	✓	✓		✓
5	The cost/benefit analysis for ending sustainment					✓	✓	✓	✓		✓
6	Market analysis/Financial analysis that supports Program exit					✓	✓	✓	✓		✓
7	Approved risk management plans, if necessary					✓	✓	✓	✓		✓
8	Consensus on program exit strategy					✓	✓	✓	✓		✓
9	Consensus on fleet retirement plan with level one schedule					✓	✓	✓	✓		✓
10	Intellectual property provisions with approval by Legal					✓	✓	✓	✓		✓
11	Consensus on trigger condition points through PMT disbanding					✓	✓	✓	✓		✓
12	The intent of the above requirements shall be met during gate review using standard subgate F template (form 1234), covering a requirement checklist (form 1234), open actions, major changes since last gated review and identified exit criteria actions					✓	✓	✓	✓		✓

* Archetype 10 is aligned with the full entry and exit criteria defined in section 5.2 of this document